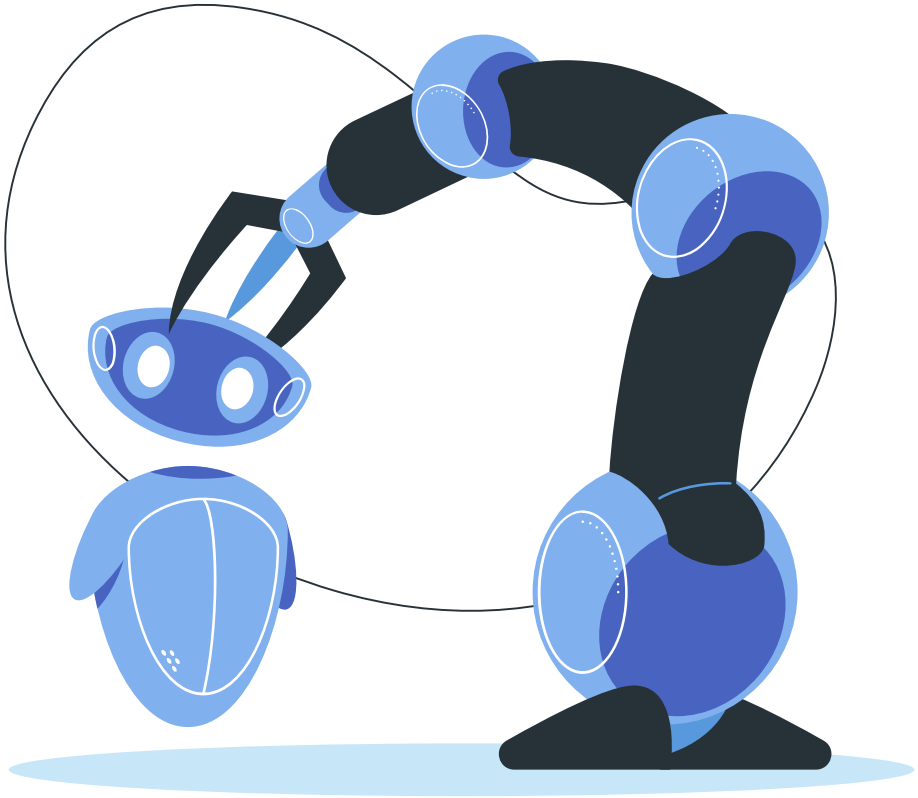
IoT

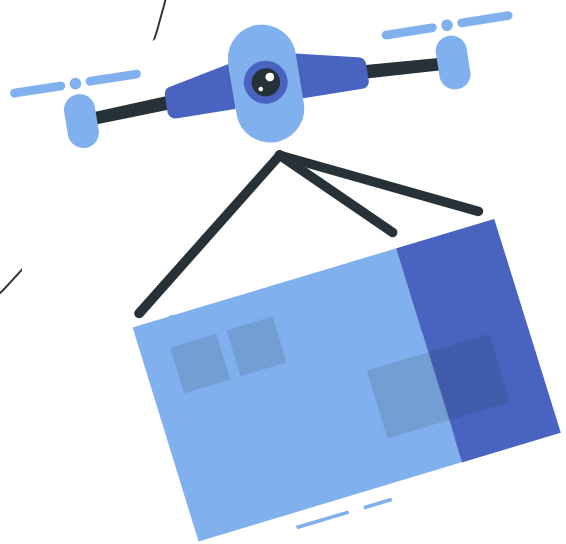
Internet of Things



What is the Internet of Things?

The Internet of Things, or IoT, is a system of **smart devices** that connect to the internet. These devices have **sensors** to collect information and **connectivity** to send or receive data.

For example, a smartwatch checks your heart rate and sends the information to your phone.
Right now, many IoT devices are helping people in their homes, offices, or cities, controlling lights, **monitoring** temperature, and saving energy.



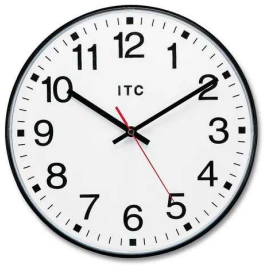
Smart or not Smart?



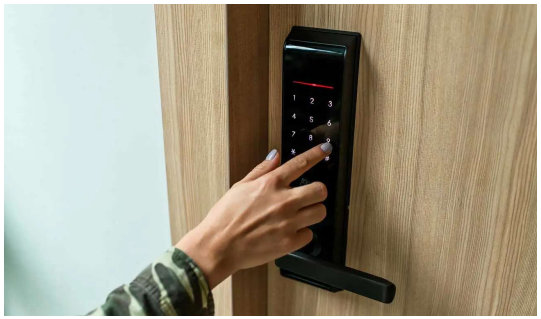




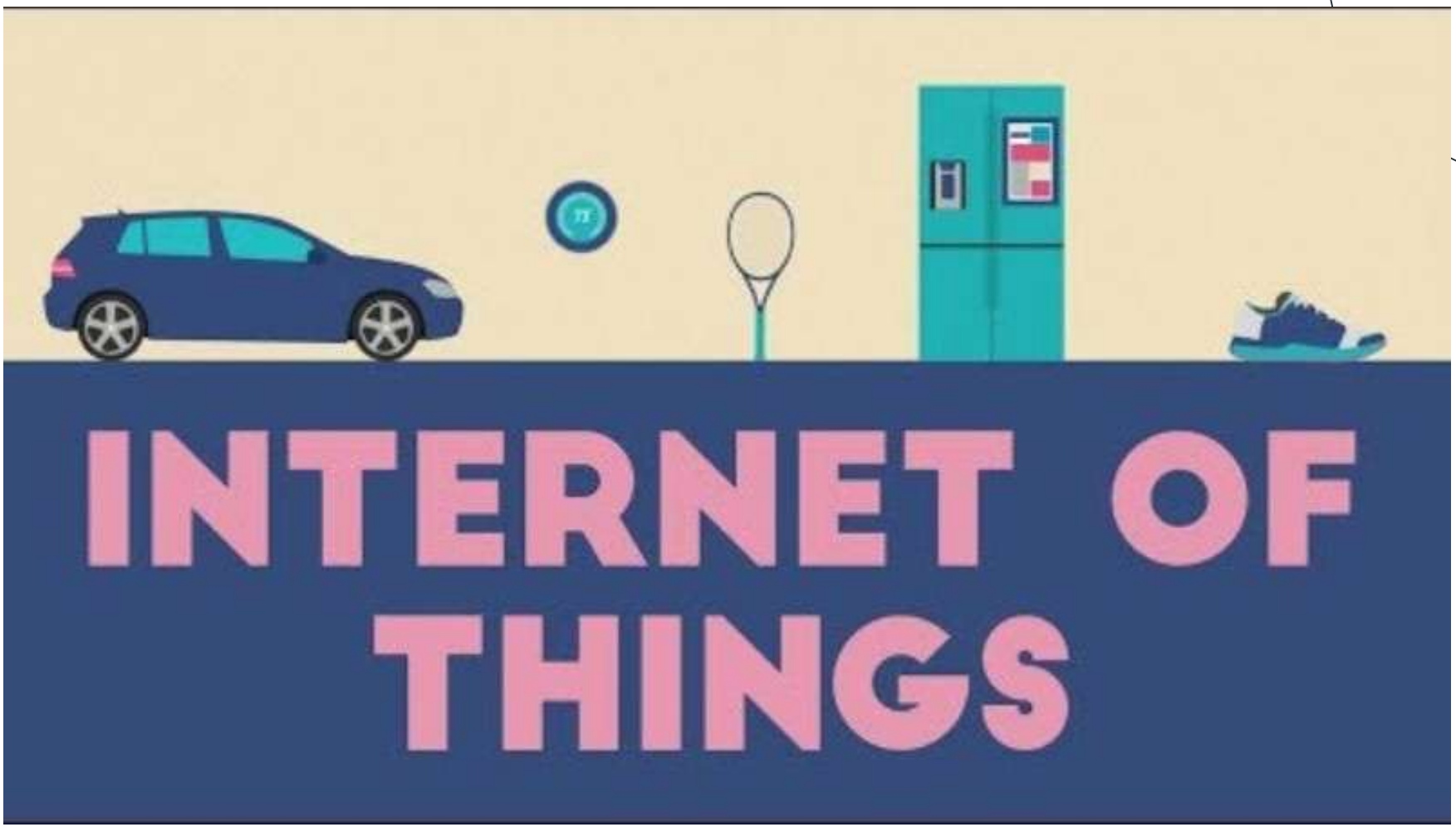











INTERNET OF
THINGS

IoT

The Internet isn't just about computers anymore. Now all kinds of everyday objects can connect to the internet, including cars, thermostats, sporting equipment, refrigerators and even shoes. This is known as The Internet of Things. Now you may be wondering why would shoes need to connect to the internet? Well the shoes can collect data like how many steps you take, which can be viewed from another Internet connected device like a smartphone. All the metrics collected by the shoes can be analyzed to tell you how many calories you've burned, giving you personalized fitness advice and much more, but the Internet of Things isn’t limited to consumer products; there are city trash cans that send alerts when they need to be emptied. Sensors on bridges that check for stress or damage to its structure and many more examples extending to Health Care, manufacturing and agriculture among others. Unfortunately there are also new security concerns, for example if you have internet connected locks a hacker might be able to unlock your front door remotely and there's the issue of privacy. All of these devices are collecting large amounts of data that can capture your actions and location throughout the day. Often we don’t exactly know all that is being collected or how this information is being used, the internet of things is going to grow exponentially over the next few years bringing with it new benefits and challenges.

Quick Comprehension Questions

1. What is the Internet of Things? Everyday objects that can connect to the internet.
2. Two examples of IoT devices from the video. Smart shoes and a smart refrigerator.
3. What kind of data can smart shoes collect? Steps and calories.
4. How is the Internet of Things expected to impact society in the future? Bringing new benefits and challenges.
5. What are the security or privacy concerns mentioned? Hackers and data collected in a day.
6. As a software developer, what is the most significant benefit of the Internet of Things for your job?

Answer



What is this?

What does it do?

How does it work?

What sensors does it use?

What is the watch doing now?

Language Focus – How to Explain a Device

Question	Useful Language	Example
1. What is it?	It is a...	It is a smart camera.
2. What does it do?	It is a device that.. It + verb (Present Simple)	It is a device that detects movement. It detects movement.
3. How does it work?	It + has / uses / connects / sends	It has motion sensors and connects to your phone by WIFI. It uses electrical energy.
4. What is it doing now?	It is + verb-ing (Present Continuous)	It is recording a video now.

Examples

PS 1. The sensor doesn’t check the air quality.

PS 2. This app shows the temperature on your phone.

PC 3. The device isn’t connecting to Wi-Fi at the moment.

PS 4. A smartwatch sends data every minute.

PC 5. Right now, the smart light is turning on.

Grammar Focus

Present Simple	Present Continuous
Describes routines, facts, functions.	*Describes actions happening now.*
It connects to Wi-Fi.	It is connecting to Wi-Fi now.
The device sends data every hour. The shoes count the steps.	The device is sending data right now. We are studying English. It is raining.
S + V (s) + C	S + to be + v + ing

Complete the sentences with your own ideas:



1. This device is _____
2. It has _____
3. It connects to the Smart phone. It sends _____
4. Right now, it is _____

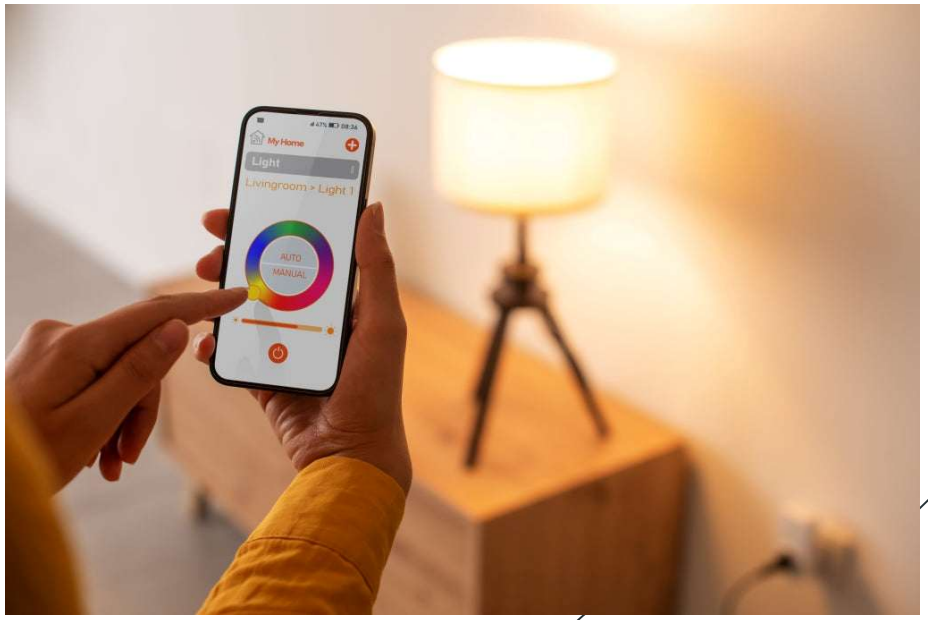
Complete the sentences with your own ideas:



1. This device is a Smart watch.
2. It has motion, wifi, bluetooth, temperature sensors.
3. It connects to the Smart phone. It sends personal data.
4. Right now, it is showing the time.

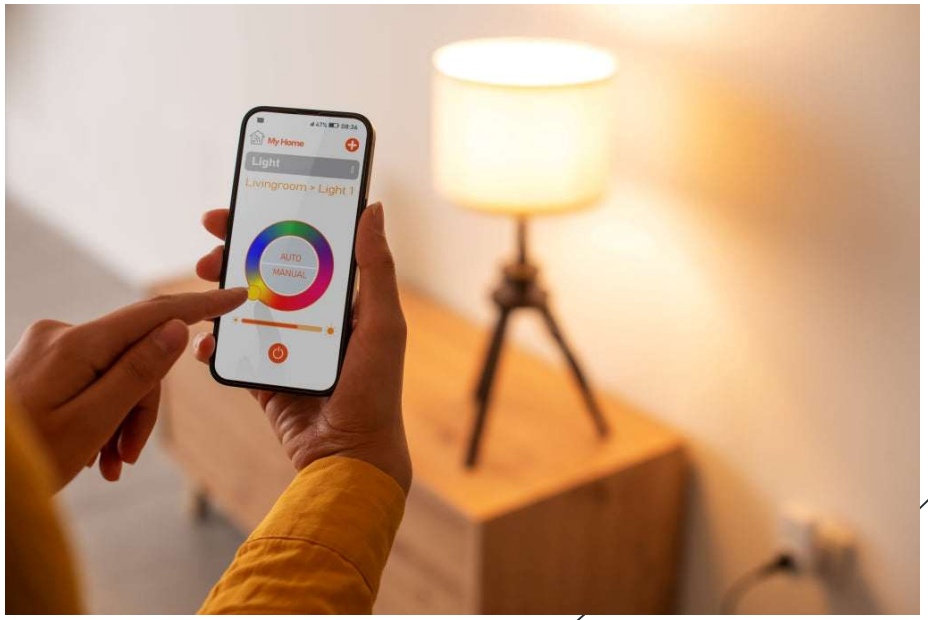
Complete the sentences with your own ideas:

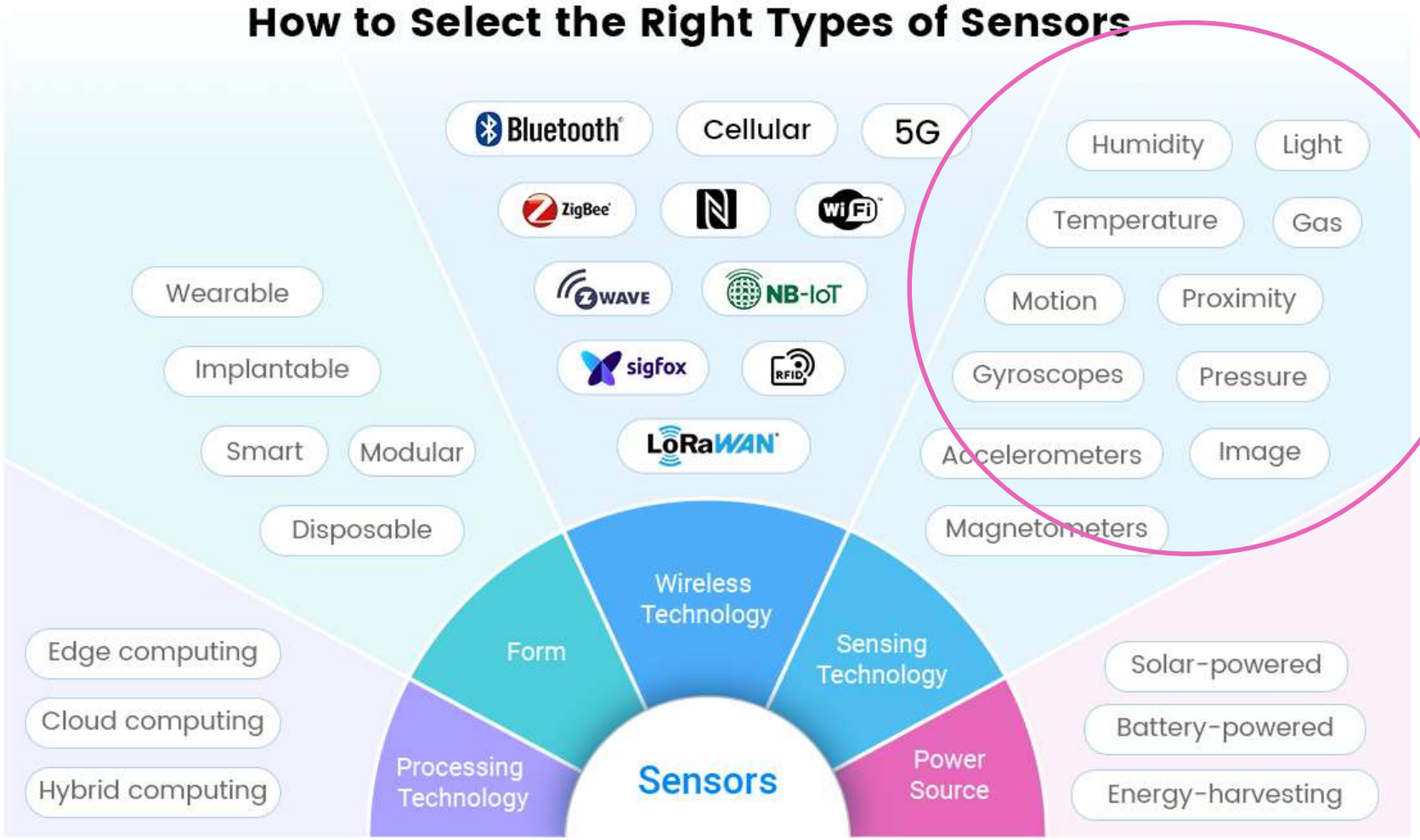
1. This device is a _____.
2. It has _____ sensors.
3. It connects to _____.
   It sends _____.
4. Right now, it is _____.



Complete the sentences with your own ideas:

1. This device is a smart light.
2. It has light and motion sensors.
3. It connects to Wi-Fi and sends data to your phone.
4. Right now, it is changing the light color.




How to Select the Right Types of Sensors
Bluetooth
Cellular
5G
ZigBee
Wi Fi
Z-WAVE
NB-IoT
sigfox
RFID
LoRaWAN
Wearable
Implantable
Smart
Modular
Disposable
Humidity
Light
Temperature
Gas
Motion
Proximity
Gyroscopes
Pressure
Accelerometers
Image
Magnetometers
Edge computing
Cloud computing
Hybrid computing
Form
Wireless Technology
Sensing Technology
Processing Technology
Sensors
Power Source
Solar-powered
Battery-powered
Energy-harvesting

Give information about these IoT devices.










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